UNIT 10 · RAPID REVISION

Formula and Fact Sheet

Precision agriculture and GNSS

Item	Value
Precision agriculture	Site-specific management by within-field variability
PA cycle	Observe → map → decide → act (VRT)
Raster vs vector data	Grid cells vs points/lines/polygons
GPS segments	Space (satellites), control (ground stations), user (receivers)
GPS satellites	≥ 24 nominal, ~31 in orbit, ~20,200 km
Positioning method	Trilateration; ≥ 4 satellites (3 for x, y, z + 1 for clock)
DOP	Dilution of precision — lower is better
Autonomous GPS accuracy	~5–10 m
DGPS accuracy	Sub-metre to ~3 m
RTK accuracy	~2–3 cm (carrier phase)
India's SBAS	GAGAN
Selective Availability	Switched off in 2000
NavIC	7 satellites (design), < 10 m over India, India + 1500 km, SPS + RS
Global GNSS	GPS (USA), GLONASS (Russia), Galileo (EU), BeiDou (China)

GIS, remote sensing and VRT

Item	Value
GIS	Capture, store, analyse and display spatial data as layers
Kriging	Interpolation of samples into a surface
Yield monitor inputs	Grain flow, moisture, header width, speed, GPS
Four RS resolutions	Spatial, spectral, temporal, radiometric
Passive vs active sensors	Reflected sunlight vs own energy (radar, LiDAR)
NDVI	(NIR - RED)/(NIR + RED); healthy crop ~0.6–0.9
Map-based vs sensor-based VRT	Prescription map + GPS vs real-time sensor
VRT control chain	GPS/sensor → controller → actuator → rate
ISOBUS	ISO 11783 CAN-bus; Task Controller = section control + variable rate
Surface analysis	Slope, aspect and flow from elevation

Precision agriculture, mapping and GPS (Topics 1–3)

- Precision agriculture = site-specific management by within-field variability; cycle observe → map → decide → act (VRT); raises input-use efficiency, yield, profit; cuts environmental impact.
- Variability: soil, yield, topographic, crop; maps → yield/soil/NDVI/prescription; management zones; data = raster vs vector; everything geo-referenced.
- GPS 3 segments: SPACE (satellites ≥ 24), CONTROL (ground master/monitor stations), USER (receivers). Position by TRILATERATION; ≥ 4 satellites (3 for x, y, z + 1 for clock); low DOP = better.
- Errors: iono/tropo delay, clock/orbit, multipath, geometry (DOP); autonomous GPS 5–10 m. Correction: DGPS (sub-m–3 m), RTK carrier phase (~2–3 cm), SBAS (GAGAN in India).
- GNSS: GPS (USA), GLONASS (Russia), Galileo (EU), BeiDou (China); regional NavIC/IRNSS (India, 7 satellites, < 10 m, SPS + Restricted) and QZSS (Japan).

GIS, sensing, VRT and environment (Topics 4–11)

- GIS = capture/store/analyse/display spatial data as LAYERS; vector vs raster; overlay, interpolation (kriging), zones, prescription maps.
- Yield monitor = grain flow + moisture + width + speed + GPS → yield map (calibrate, correct flow lag) — the start of the PA cycle.
- Remote sensing: 4 resolutions — SPATIAL (pixel), SPECTRAL (bands), TEMPORAL (revisit), RADIOMETRIC (levels); passive (sunlight) vs active (radar/LiDAR); satellite/aerial/drone.
- NDVI = (NIR - RED)/(NIR + RED); healthy crop 0.6–0.9; also EVI, SAVI, LAI; proximal N-sensors for sensor-based VRT.
- VRT: MAP-based (prescription + GPS) vs SENSOR-based (real-time); GPS/sensor → controller → actuator; ISOBUS (ISO 11783) Task Controller = section control + variable rate; RTK auto-steer.
- PA cuts run-off, GHG, energy, compaction and water use; barriers = cost, data, skill (CHC/drone-as-a-service).

Further reading

- Srinivasan — Handbook of Precision Agriculture: Principles and Applications.
- Lillesand, Kiefer & Chipman — Remote Sensing and Image Interpretation.
- NovAtel — An Introduction to GNSS.
- ISO 11783 (ISOBUS) — tractor–implement communication.
- ISRO NavIC and GAGAN programme pages (current constellation status).
